

# RANBIR KUMAR

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## EDUCATION

**B.Tech, Computer Science (AI & ML)** — Vellore Institute of Technology, Chennai | CGPA: 8.28/10 | Aug 2023 – May 2027  
**Coursework:** Data Structures & Algorithms, Machine Learning, Deep Learning, Computer Vision, Explainable AI, DBMS & SQL

## EXPERIENCE

**Software Testing & Product Development Intern** | **IAMAI** (Internet & Mobile Association of India) | May 2026 – Jul 2026

- Automated manual pre-release QA with an **LLM-planned browser agent** (Playwright, Chromium) that reads DOM and accessibility snapshots, executes safe UI actions, and emits machine-readable defect reports, cutting click-through effort an estimated 70% per release.
- Ran k6 load, stress, and spike profiles plus Burp Suite scans, mapping staging degradation thresholds pre-launch.

**Technical Integration Lead, AI Systems & Embedded Products** | **IORA**, Noida | May 2025 – Apr 2026

- Drove architecture reviews across PCB, firmware (MQTT over TLS, GNSS, flash-buffered telemetry), and power subsystems for an nRF9151 LTE-M/NB-IoT wearable, aligning 3 teams on one interface contract.
- Architected the company's agentic RAG platform on FastAPI and Next.js, adding response caching and failover across 3 LLM providers so a vendor outage degraded gracefully.

**Deep Learning & Computer Vision Intern** | **Proeffico Solutions Pvt. Ltd.**, Noida | May 2025 – Oct 2025

- Shipped 4 production vision pipelines (speed, pothole, fire, workplace safety) on Jetson Xavier NX over live RTSP feeds at sub-40 ms latency, roughly 25 FPS per stream, replacing continuous manual review.

**Rover Lead, Autonomous Systems & Computer Vision** | **Technocrats Robotics**, VIT Chennai | Aug 2025 – Aug 2026

- Led a 60-member Rover Department, owning subsystem architecture, code review, and the build timeline for a perception stack spanning soil classification, arrow-marker detection, and an end-effector camera pipeline.

## PROJECTS

**IORA DocQA** — Production Agentic RAG Document Platform | *FastAPI · Next.js · pgvector · DuckDB · Multi-LLM*

- Engineered ingestion, covering parsing, chunking, and embedding into a Supabase pgvector store, behind a FastAPI retrieval service with a hard coverage gate that refuses unsupported answers.
- Extended it with a DuckDB text-to-SQL path over uploaded CSV and XLSX, per-user memory, and a semantic cache resolving an estimated 25% of repeat queries without an LLM call.

**IAMAI Publishing Platform** — Full-Stack CMS for a Live Public Website | *Next.js · Supabase · Postgres · GitHub Actions*

- Modelled eleven content collections on a normalized Postgres schema with CMS-managed page blocks driving every public page, guarded by RBAC and released through GitHub Actions to Netlify under a smoke-test monitor.

**TerraView** — Indoor Venue Navigation & Routing Engine | *React · Vite · Graph Pathfinding* [jio-map.netlify.app](https://jio-map.netlify.app)

- Represented a convention centre as a single routing graph over stall, gate, amenity, and lift nodes, resolving cross-hall and cross-floor journeys across 4 floor views and indexing 393 stalls and 225 companies into an offline zero-backend bundle embedded in Android and iOS WebViews at Global Fintech Fest.

**RetinaProto** — Few-Shot Medical Image Classifier with Grad-CAM | *PyTorch · Prototypical Networks · ResNet-18*

- Implemented Prototypical Networks from scratch (episodic N-way K-shot sampling, class prototypes, Euclidean inference) over an end-to-end ResNet-18, reaching 93% accuracy on hospital retinal images with per-prediction Grad-CAM attribution.

## RESEARCH & PATENTS

**Patrolling Bot** — Autonomous Campus Surveillance Robot | *Patent filed* — YOLO detection (missing ID cards, fire/smoke, overcrowding) and action recognition for fight detection, alerting security in real time from Jetson Xavier NX with LiDAR/ultrasonic navigation, built in Python with OpenCV, PyTorch, and ROS.

**Vision-Based Terrain Analysis** | *Accepted; publication forthcoming* — YOLOv8 fused with MiDaS monocular depth for single-camera terrain classification on embedded edge hardware.

## SKILLS

**Languages:** Python, TypeScript, JavaScript, Java, SQL

**AI & Machine Learning:** PyTorch, TensorFlow, scikit-learn, YOLOv8, MiDaS, OpenCV, ResNet, Prototypical Networks, Grad-CAM, SHAP, LIME, Jetson edge inference

**Generative & Agentic AI:** Agentic RAG, AI Agents, LangGraph, Tool Calling, Model Context Protocol, Vector Search (pgvector, FAISS), LangChain, Semantic Caching, Text-to-SQL

**Backend & Data:** FastAPI, Flask, Next.js, React, REST API Design, PostgreSQL, Supabase, DuckDB, Redis, RBAC & RLS, System Design

**Cloud & Tooling:** Docker, Git, GitHub Actions, Linux, Vercel, Netlify, Railway, Playwright, pytest, k6

## ACHIEVEMENTS

Patent Filed · 2-time Finalist, DD Robocon & IRC · IRC Top 20 Worldwide 2025 · National Finalist, SIH 2025 & Insightify 6.0